CSE 4241: Bioinformatics

Assignment on Bioinformatics

A Comparative Analysis of Two Research Papers

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Papers Analysed:

1. *An Interpretable Machine Learning Approach to Identify Mechanism of Action of Antibiotics* (**Paper 1** / InterPred)
2. *Advances in Estimating Level-1 Phylogenetic Networks from Unrooted SNPs*(**Paper 2** / CUPNS)

Part 1 — Understanding the Papers

Paper 1: InterPred — Interpretable ML for Antibiotic Mechanism of Action

1. Problem and importance. Antibiotic resistance has quietly become one of the most serious threats in modern medicine. Drug-resistant infections already kill around 700,000 people every year, and if the trend continues unchecked, projections put that number above ten million by 2050—a figure that would place it ahead of most cancers as a cause of death. Finding new antibiotics is therefore urgent, but the discovery pipeline has a specific bottleneck that this paper tries to address.

Machine-learning tools like the Stokes et al. model (which discovered Halicin) are very good at scanning large compound libraries and flagging bioactive molecules. The problem is that they spit out hundreds or even thousands of hits with no explanation of *how* the compounds work. Figuring out the mechanism of action (MOA) experimentally is slow and expensive, so you cannot realistically chase every candidate. InterPred is built to fix that: it makes the prediction interpretable enough that a researcher can get a reasonable MOA guess at the same time as the activity prediction, without needing extra wet-lab work just to sort the list.

2. Main idea and method. Rather than using a graph neural network that no one can look inside, InterPred breaks every molecule down into a fixed set of recognisable structural pieces—ring systems and functional groups—using RDKit. Each molecule becomes a binary vector ("does it contain a β-lactam ring? yes/no"). An L1-regularised logistic regression or an Extra-Trees classifier is then trained on these vectors. Because every feature maps directly onto a named chemical fragment, the model's own coefficients tell you which piece of the molecule is driving the prediction—no separate explanation step needed.

A companion tool called *MOACluster* takes this further: it groups molecules that share the same high-weight moieties, and the paper shows that compounds landing in the same cluster tend to share a mechanism of action in practice.

3. Key assumptions.

- Bioactivity is mostly determined by local structural fragments (rings and functional groups) rather than the molecule's overall three-dimensional shape or conformation.
- Molecules that share a dominant moiety tend to act through the same mechanism.
- The active/inactive training labels are reliable enough to learn from.
- A shallow, linear model over this fixed feature set can come close to matching a deep graph network's predictive accuracy.

4. Key limitations. The fixed feature dictionary is also the model's main blind spot. If a genuinely novel scaffold drives activity, but that scaffold does not appear in the predefined catalogue of rings and functional groups, the model simply cannot see it—which is ironic, since finding novel mechanisms is exactly what the paper is trying to do. Separately, a statistical association between a moiety and an MOA class is not the same as a causal link; with a small or imbalanced dataset those correlations can be spurious. The experimental validation covers only five moiety–MOA pairs, which is an illustration rather than a thorough evaluation. And the benchmark shows a meaningful fraction of test molecules are structurally very close to training compounds, which can inflate the reported accuracy on truly novel chemistry.

Paper 2: CUPNS — Phylogenetic Networks from Unrooted SNPs

1. Problem and importance. Standard phylogenetic trees assume that lineages only ever split—they never merge. In practice, evolution is messier than that. Recombination, hybridisation, and horizontal gene transfer all cause lineages to mix back together, and a tree simply cannot represent those events. Phylogenetic *networks* can, but unconstrained networks are too complex to recover reliably from data. This paper focuses on *level-1* networks—sometimes called galled trees—where the reticulate cycles do not overlap with each other.

The specific question the authors tackle is whether you can reconstruct such a network using only SNP data, without knowing which allele state is ancestral. Ancestral state is usually unavailable in real biological data, so earlier methods that required it were not truly practical. Prior work either depended on that information or could not guarantee correctness. CUPNS is designed to solve both problems at once.

2. Main idea and method. The paper first establishes, mathematically, that the GBP algorithm cannot work in this setting. SNP data without ancestral state simply cannot supply the full set of quartet trees that GBP requires—it is a structural impossibility, not just a practical difficulty.

CUPNS then takes a different route. It constructs quartet trees from the available SNPs, builds an intermediate structure called an *SN-tree*, and resolves the ambiguous multi-branching nodes (polytomies) into cycles. The authors prove that this procedure recovers the correct network provided two conditions hold: every cycle in the true network has at least five nodes, and the SNP data covers every bipartition of the network. As a bonus, they also supply the first formal correctness proof for Gusfield's (2005) algorithm, which had been used in practice but never rigorously justified.

3. Key assumptions.

- Every reticulation cycle in the true network has length ≥ 5. Shorter cycles are provably unidentifiable, so this bound is not a convenience—it is mathematically necessary.
- SNPs are biallelic and show no homoplasy (no site mutates independently more than once).
- A perfect Oracle correctly identifies every true SNP site, with neither false positives nor false negatives.
- The network is level-1 (non-overlapping cycles).
- The available SNPs cover every bipartition in the true network.

4. Key limitations. The gap between the theory and real biological data is substantial, and the authors are honest about it. The most critical fragility is the Oracle assumption: even a single false-positive SNP call—one homoplastic site incorrectly accepted as a clean SNP—can introduce a quartet that contradicts the true topology, and that one bad quartet can derail the entire reconstruction. False negatives (missed SNPs) are tolerable as long as enough coverage remains, but false positives are genuinely dangerous. CUPNS is also computationally heavier than Gusfield's approach. The whole framework applies only to level-1 networks; more complex reticulation structures break identifiability. The authors themselves describe the work as primarily relevant from a theoretical standpoint for now.

Part 2 — Comparing the Two Papers

1. Which paper is more practical for real-world use?

Paper 1 is clearly more ready to deploy. InterPred was trained and evaluated on real screening data—the CO-ADD dataset, the FDA approved library, and the Drug Repurposing Hub—and it produces output a bench chemist can directly act on: a ranked list of moieties driving bioactivity and a set of MOA-consistent clusters. The tools it depends on (RDKit, scikit-learn) are standard and freely available. You could run it on a new hit list today.

CUPNS requires a perfect Oracle—something no real sequencing pipeline can offer. The authors are upfront that the method is positioned as theoretical groundwork rather than a deployment-ready tool. Until error-tolerant extensions are built on top of it, the gap between theory and practice remains wide.

2. Which paper is more theoretically rigorous?

Paper 2 is the stronger of the two on formal grounds, and it is not particularly close. The paper contains actual mathematical proofs: network identifiability, algorithmic correctness, statistical consistency, and closed-form sample-complexity bounds. These are guarantees in the strict sense—they hold unconditionally given the stated assumptions.

InterPred's justification is empirical: cross-validated AUC, ROC curves, and a few illustrative examples. That is perfectly appropriate for an applied ML paper, but it demonstrates that the method *works on the datasets tested*, not that it *must* work in general. Whether the fixed feature set is sufficient, or whether a large coefficient genuinely indicates the causal moiety rather than a correlated one, is left without a formal answer.

3. Which method is more sensitive to data quality?

Paper 2's methods are more fragile, and this fragility is not a vague concern—it is something the proofs themselves establish. A single false-positive SNP can cause the reconstruction to fail completely, rather than degrade gradually. InterPred is more forgiving in comparison: mislabelled activity calls or class imbalance erode accuracy incrementally but the model keeps producing output. In a practical setting, a method that gives approximate answers under imperfect conditions is usually far more useful than one that silently fails.

Summary comparison table

Dimension	Paper 1 — InterPred	Paper 2 — CUPNS
Nature of work	Empirical ML pipeline with interpretable features	Formal algorithms with mathematical proofs
Evidence base	Real assay data (FDA library, CO-ADD screen)	Idealised assumptions; no empirical validation
Deployment readiness	High — usable on current screening hit lists	Low — authors frame it as theoretical groundwork
Formal guarantees	None beyond cross-validated AUC	Correctness, uniqueness, consistency, sample bounds
Failure under noise	Gradual accuracy loss; output still produced	Can fail outright from a single bad SNP call
Data sensitivity	Moderate; class imbalance is the main risk	Very high; false-positive SNPs are catastrophic
Primary output	Ranked moiety list and MOA clusters	Reconstructed semi-directed network topology
Practical value	Direct triage of high-throughput screening hits	Foundation for future applied network tools

Part 3 — Brainstorming and Critical Thinking

1. One way to combine ideas from both papers

Both papers are, at their core, trying to recover a hidden structure from noisy, binary data while keeping the result either interpretable or provably correct. That shared framing suggests a natural bridge. **Paper 2** assumes a perfect Oracle but says nothing about how one would be built in practice. **Paper 1** is essentially a worked example of building an interpretable classifier from hand-chosen features.

Putting the two together: instead of assuming a flawless SNP detector, you could train an InterPred-style logistic regression over column-level features of a DNA alignment—minor-allele frequency, pairwise linkage statistics, compatibility with neighbouring sites—to produce a probability that each column is a genuine SNP rather than a homoplastic artifact. Because the resulting model is transparent, a biologist can inspect which features pushed a borderline column toward acceptance or rejection. Those soft probability estimates could then feed CUPNS as confidence-weighted quartet votes instead of hard binary calls, converting an unrealisable idealisation in **Paper 2** into something learnable and inspectable.

2. What happens when data is incomplete or noisy

For InterPred (Paper 1). With limited or mislabelled data, the L1 model can latch onto a moiety simply because it happened to co-occur with activity a few times in the small positive class—not because it is actually responsible. The 95% class imbalance makes this worse: the minority class has so few examples that even a single unlucky split can look meaningful.

A practical fix: Apply *stability selection*. Train the model repeatedly on random subsamples of the data and report only those moieties that appear as significant features consistently across the majority of runs. A moiety that survives resampling is far more likely to be real than one that appeared in a single lucky split.

For CUPNS / Gusfield (Paper 2). Incomplete SNP coverage leaves polytomies in the SN-tree that cannot be closed into cycles—there simply is not enough information to distinguish which topology is correct. And as noted above, even one false-positive SNP can corrupt the reconstruction entirely.

A practical fix: Borrow the logic of bootstrap support from species-tree estimation. Subsample the available SNPs repeatedly, run CUPNS on each subsample, and track how consistently each polytomy resolves. Polytomies that resolve differently across runs would be flagged as low-confidence rather than silently included in the final answer. This converts the current all-or-nothing guarantee into a graded, uncertainty-aware output.

3. A new idea inspired by both papers

Both papers encode biological information as binary profiles—moiety presence/absence in **Paper 1**, SNP states across taxa in **Paper 2**. That parallel structure suggests a novel application: building a *compound bioactivity network*.

For each antibiotic candidate, InterPred's moiety-based features can be used to construct a "bioactivity fingerprint" across a panel of pathogens or assay conditions—essentially the same kind of binary matrix that CUPNS takes as input, but over compounds rather than taxa. Running a CUPNS-style network inference on these profiles would cluster compounds with similar activity patterns as tree-like branches, while compounds that are partially active across two mechanistically distinct classes would appear as reticulate (hybrid) nodes.

This is biologically meaningful beyond just being a clever analogy: antibiotics that engage two independent targets simultaneously are much harder for bacteria to develop resistance against, because a single resistance mutation would need to neutralise both mechanisms at once. A compound showing up as a hybrid node in this network would be a high-priority candidate for follow-up, flagged automatically by the structure of the data rather than by manual inspection.

Part 4 — Application

*(Option A: How can **Paper 1** help in discovering new antibiotics?)*

High-throughput screening can now test millions of compounds in a short time, but finding hits is no longer the hard part—deciding which ones are worth following up is. A single large screen can return hundreds of bioactive molecules, and characterising each one experimentally (resistance-frequency assays, target identification, mechanism confirmation) is expensive enough that most of them will simply never get looked at. InterPred inserts a fast, cheap triage step between the initial screen and the laboratory that helps make that prioritisation principled rather than arbitrary.

The logic is straightforward. If a hit's dominant structural moiety matches one already associated with a well-understood mechanism—a β-lactam ring pointing to cell-wall synthesis inhibition, for instance—it can be deprioritised unless there is some other compelling reason to investigate it. Discovering another variant of a mechanism we already have many drugs for is rarely the most valuable use of lab time. On the other hand, a compound flagged as strongly bioactive whose key moiety does not match any existing MOA cluster is exactly the kind of molecule worth spending resources on: it may operate through a target that bacteria have had no evolutionary exposure to, meaning resistance is less likely to be pre-existing or easily acquired.

MOACluster strengthens this further by grouping molecules that share a mechanistic signature regardless of their overall scaffold. Two compounds from entirely different chemical families landing in the same cluster points to the shared moiety—not the scaffold—as the driver of activity. That is a useful hint for medicinal chemists trying to strip a complex natural-product hit down to its pharmacologically essential core. Taken together, InterPred and MOACluster convert an unwieldy hit list into a structured set of priorities, with the reasoning behind each priority made explicit in the model's own coefficients.

Part 5 — Reflection

1. Which paper was harder to understand?

Paper 2 was considerably more demanding. It requires holding a lot of specialised vocabulary in mind simultaneously—level-1 networks, galled trees, quartets versus bipartitions versus clades, semi-directed topologies, the distinction between $Q(N)$ and $Q_r(N)$—before the proofs even begin. And the proofs themselves are multi-case combinatorial arguments where the logic in one lemma feeds directly into the next theorem, so losing the thread early makes everything downstream harder to follow.

Paper 1 follows a much more familiar structure: motivate the problem, describe the features, train a model, evaluate with standard metrics. The biology is non-trivial, but logistic regression, ROC/AUC analysis, and feature importance are tools most computer science students encounter relatively early, which made it easier to engage with the paper's technical content on a first read.

2. Which is more useful for learning?

Paper 1 is more immediately transferable as a set of skills. Interpretable feature engineering, handling severe class imbalance, evaluating a classifier on real biological data, and communicating results to domain experts who do not have an ML background—these are recurring challenges across bioinformatics and data science more broadly. The paper is also a good example of how to make a deliberate simplicity trade-off: accepting a marginal drop in accuracy (AUC 0.87 versus 0.88 for the black-box baseline) in exchange for a model whose outputs can be explained and checked.

Paper 2 is useful in a different way. It is a clean demonstration of how to think carefully about *why* an algorithm should be trusted rather than just whether it performs well on a benchmark. The process of defining what the data can and cannot identify in principle, then proving correctness under those bounds, is a template worth understanding even if the specific domain (phylogenetic network reconstruction) is narrow.

3. One suggested improvement per paper

For **Paper 1**, the weakest part of the paper is its validation: five moiety–MOA case studies that "look right" is an illustration, not a test. A straightforward improvement would be to apply a permutation test to each proposed moiety–MOA link—randomly shuffle the class labels many times and check whether the observed association appears significantly more often than chance would predict. That would give every reported link a p-value, making it easy for readers to judge which ones are robust and which are marginal.

For **Paper 2**, the most conspicuous gap is the complete absence of any simulation study. Every result is a proof under idealised conditions, which is valuable, but it leaves a large practical question unanswered: how badly does the method degrade as the Oracle error rate increases, or as SNP coverage decreases? Even a modest simulation showing performance as a function of those two parameters would make the paper much more useful to anyone thinking about building on top of these results, because that degraded regime is exactly what real sequencing data looks like.